

RESUME

VINAY M. MADGI

Hubballi, Karnataka, India

+91 7975200870 | vinaymadgi28@gmail.com

[LinkedIn](#) | [GitHub](#)



Career Objective

Motivated Computer Science undergraduate at KLE Technological University with strong foundations in Artificial Intelligence, Machine Learning, Deep Learning, Generative AI and Backend Development. Passionate about building intelligent systems, solving real-world problems, and contributing to innovative software engineering teams. Seeking AI/ML Engineer, DL Engineer, Gen AI Engineer, Data Engineer, Backend Developer or Software Development Engineer (SDE) opportunities.

Education

Examination Passed	University / Board	Year of Passing	CGPA / Percentage
B. E. (CSE)	KLE Technological University, Hubballi	2023-27	8.34 (1 st to 6 th semester)
PUC	Pre-University Board, Bengaluru	2023	70
Class-X	Central Board of Secondary Education	2021	68

Technical Skills

- Programming Languages: Python, Java, SQL, C++
- Web & Database Technologies: Django, MySQL, MongoDB
- AI/ML & Data Science: PyTorch, LangChain, FAISS, Transformers, Retrieval-Augmented Generation (RAG), Meta-Learning.
- DevOps & Cloud: Git, GitHub, Postman, Docker, Kubernetes, Prometheus, Grafana
- Core Subjects: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems.

Projects

Autonomous Citation-Grounded Academic Report Generation via GraphRAG and NLI-Driven Reflection.

Technologies: *Python, PyTorch, LangChain, Streamlit, FAISS, Transformers*

- Developed an end-to-end AI-powered academic report generation system using LLMs, Retrieval-Augmented Generation (RAG), and vector databases.
- Implemented hierarchical planning and self-reflection mechanisms to improve factual accuracy and citation coverage.
- Integrated DeBERTa-MNLI and SciFact-based verification for hallucination detection and evidence validation
- Built a Streamlit web application with report management, citation handling, and document export features.

Amortized Hyperparameter Optimization for Few-Shot Image Classification

Technologies: *Python, PyTorch, Prototypical Networks*

- Developed a meta-learning framework for N-way K-shot image classification.
- Created a dynamic episodic task generation pipeline for Omniglot and CIFAR-100 datasets.
- Designed CNN-based embedding networks and evaluated multiple optimization strategies.
- Conducted automated experiment tracking and comparative performance analysis.

Sports Inventory Management System

Technologies: *React.js, Node.js, Express.js, MongoDB*

- Built a full-stack MERN application for real-time sports inventory tracking and management.
- Developed REST APIs for inventory operations and reporting.
- Designed responsive dashboards and user-friendly interfaces.
- Implemented scalable database architecture and CRUD functionalities

Publication in a Journal

Parameter-Efficient Adaptation of Vision-Language Models for Domain-Specific Multimedia Retrieval in Fashion.

- **Communicated** to Multimedia Tools and Applications, Springer.
- **Under Review** (Submitted Reviewers' Response)

Amortized Hyperparameter Optimization: A Static Meta-Learning Approach for Few-Shot Image Classification.

- **Communicated** to Neorocomputing, Elsevier.

Publication in Conference

An Edge-Native IoT Smart Surveillance System with Multi-Factor Threat Reasoning, 8th International Conference on Computational Intelligence in Communications and Business Analytics (CICBA-2026), organized by the Department of Computer Science, University of Gour Banga, Malda, India, from 10–12 September, 2026.

- **Accepted** for presentation and publication in Communications in Computer and Information Science, Springer Nature. (**SCOPUS**)

Efficient Image-Text Alignment for Fashion Data Using Frozen Vision-Language Encoders, 5th International Advanced Computing and Intelligent Technologies (ICACIT 2026), jointly organized by Netaji Subhas University of Technology, East Campus, Delhi, and Indira Gandhi National Tribal University, RCM-Manipur, Imphal, 27-28 March 2026.

- **Presented & Accepted** for publication in Lecture Notes in Electrical Engineering, Springer Nature. (**SCOPUS**)

Analysis of Factors Causing Infant Mortality in Karnataka and Recommendations for Prevention, 2025 International Conference on Intelligent Systems and Pioneering Innovations in Robotics and Electric Mobility (INSPIRE), Mangalore, India, 2025, pp. 139-144,

- **Published** in IEEE Xplore, doi: 10.1109/INSPIRE67328.2025.11300582. (**SCOPUS**)

Certifications

- Jenkins for DevOps: Jenkins Configuration for DevOps – Infosys Springboard (2025)
- DevOps Mindset: Understanding DevOps Tools – Infosys Springboard (2025)

Achievements

- Published international conference research paper in AI and Vision-Language Models.
- Developed advanced AI systems involving RAG, LLMs, and fact-verification pipelines
- Active involvement in research and software development.

Area Interest

- Artificial Intelligence & Machine Learning
- Generative AI and Large Language Models
- Data Engineering & Analytics

- Software Development Engineering (SDE)

Declaration

I hereby declare that the information provided above is true to the best of my knowledge and belief.

Date: 22/08/2026

Signature